Deciding the $\forall\mathsf{PO}$-Free Fragment of $\mathcal{ALCI}_{\mathrm{RCC5}}$

Ordered-Disjoint Models, Cone Schemes, and a Proof Checked in Lean 4

Michael Wessel
LambdaMikel@Home • miacwess@gmail.com

with AI assistants:
Claude Opus 4.6, Opus 4.7, Opus 4.8, Fable 5, Opus 5 (Anthropic)
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Abstract

$\mathcal{ALCI}_{\mathrm{RCC5}}$ is the description logic $\mathcal{ALCI}$ whose roles are the five RCC5 relations—equality, proper part, proper superpart, partial overlap and disjointness—read as a total, converse-coherent labelling of pairs of elements that obeys the RCC5 composition table, with equality as identity. Whether concept satisfiability in this logic is decidable has been open since Wessel's report of 2002/2003. We prove that it is decidable for the fragment in which negation normal form contains no universal restriction over partial overlap. Everything else stays: existential overlap, both quantifiers over disjointness and over both directions of proper part, and arbitrary nesting; the fragment has no finite model property. The decision procedure computes a greatest fixed point over a finite space of *cone signatures*, each a Hintikka-style type paired with a set of types permitted strictly below it, and accepts a concept when a surviving signature contains it. That every satisfiable concept is accepted holds for the whole logic and is proved by reading signatures off an arbitrary model. That every accepted concept of the fragment is satisfiable is proved by unfolding the surviving signatures into a split forest of fresh occurrences: its part-of births generate a strict order, its disjointness births a downward-closed disjointness relation, and every remaining pair overlaps. The construction rests on an ordered-disjoint normal form that characterizes RCC5 frames exactly. The whole argument—the normal form, both directions, normalization of raw formulas, and the equivalence of the abstract semantics with a semantics of plain sets—is formalized in Lean 4 as an executable decision procedure with no `sorry` and fragment membership as its only hypothesis. By a representation theorem of Lutz and Wolter the result also holds for substructure semantics over regular closed regions of $\mathbb{R}^n$. An October 2026 revision adds a certified extension, following an external review by Astra 6: concepts that do contain $\forall\mathsf{PO}$ are also decided, provided $\exists\mathsf{PP}$, $\exists\mathsf{PO}$ and $\forall\mathsf{DR}$ are absent, by erasing each $\forall\mathsf{PO}$ restriction — the first decidable slice containing $\forall\mathsf{PO}$. The search space is doubly exponential, so the contribution is a decidability theorem, not a practical reasoner.

Keywords: description logics; $\mathcal{ALCI}$; RCC5; region connection calculus; qualitative spatial reasoning; decidability; greatest fixed points; interactive theorem proving (Lean).

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1 Introduction

A description logic with spatial roles lets one say that every proper part of a region is a forest, that some region overlapping it is a river, and that nothing disjoint from it is German territory. Taking the RCC5 relations as the roles of $\mathcal{ALCI}$ gives exactly that vocabulary [2, 14]: for two regions, exactly one of EQ (equal), PP (proper part), PPI (proper superpart), PO (partial overlap) and DR (disjoint) holds. The central reasoning service, classification, reduces to concept satisfiability, and whether that problem is decidable for $\mathcal{ALCI}_{\mathrm{RCC5}}$ was left open when the logic was introduced [14]. Lutz and Wolter, who studied the corresponding modal logics systematically, named it "perhaps the most interesting candidate" for a decidable and still useful spatial logic [6], and proved that the neighbouring logics they could settle are undecidable.

Why the problem is hard. Two features combine. First, *totality*: every two elements of a model stand in some RCC5 relation, and every triple must respect the composition table. A witness introduced for one requirement is therefore immediately constrained against every other element,

not only against the element that requested it. Second, the logic has no finite model property: $\exists \mathsf{PP}.\top \sqcap \forall \mathsf{PP}.\exists \mathsf{PP}.\top$ forces an infinite chain of proper parts (Example 2.4). A decision procedure must therefore represent infinite models finitely, and the familiar way to do that—stop expanding an occurrence when an earlier occurrence carries the same type, and let the earlier one stand in for it—is exactly what fails here. Reusing an element imposes *global* conditions on the structure being built: the reused edges must not close a cycle in the part-of order, and the relations they force must be jointly realizable. A test that compares types, or that inspects one edge or one node at a time, cannot see either condition (§10).

The fragment. We forbid one construct: after negation normalization, no concept of the form $\forall \mathsf{PO}.D$ may occur. On formulas with unrestricted negation this condition is polarity-sensitive—no $\forall \mathsf{PO}$ in positive and no $\exists \mathsf{PO}$ in negative position—because $\neg \exists \mathsf{PO}.D$ normalizes to $\forall \mathsf{PO}.\neg D$. The cut is narrower than it may look: PO remains in the language, $\exists \mathsf{PO}.C$ is unrestricted, and all four quantifiers over PP, PPI and DR remain, as does the failure of the finite model property. The reason this boundary is the right one is simple to state. Whenever a composition entry offers several relations for a pair, it offers PO among them, and putting such a pair at PO costs nothing—unless some formula objects to a PO edge. The only formula that can is $\forall \mathsf{PO}.D$. In the fragment, the maximally overlapping completion of whatever the existential demands force is always available, and the proof below is organized around that fact.

Where the construction comes from. The model built in the soundness proof has the shape of Wessel's *split-forest* representation: the part-of order as a forest of trees, in which no region is shared between two demands, with every relation between branches horizontal and completed afterwards. Earlier attempts in this project tried to extract such a forest from an arbitrary model and to control the horizontal relations that cross between its branches. The cone scheme builds the forest directly instead, and on the fragment it never has to control those relations at all. Section 6.1 explains the idea and the relationship.

Contributions.

1. An exact normal form for RCC5 frames with strong equality: *ordered-disjoint structures*, a strict partial order for PP together with a symmetric, irreflexive, downward-closed disjointness for DR, with PO as the residue; and an explicit representation of every such structure by nonempty sets (§3).

2. A finite *cone scheme*—signatures, static admissibility, transition compatibility, and a monotone reductive elimination—whose greatest fixed point decides satisfiability on the fragment (§4).

3. A completeness proof that needs no fragment hypothesis: the signatures realized by any model survive their own elimination (§5). It yields, as a corollary, a sound refutation test for the whole logic.

4. A soundness proof by a *fresh-occurrence* unfolding in which no element is reused—a split forest, built rather than extracted—so that the frame conditions that defeated reuse-based constructions hold by construction (§6, §6.1).

5. Decidability for normalized concepts, for raw formulas with negation, for semantics over plain sets, and—by Lutz and Wolter's representation theorem—for substructure semantics over regular closed regions of $\mathbb{R}^n$ (§7).

$\mathsf{comp}(R,S)$	EQ	PP	PPI	PO	DR
EQ	{EQ}	{PP}	{PPI}	{PO}	{DR}
PP	{PP}	{PP}	$\mathcal{U}$	{PP, PO, DR}	{DR}
PPI	{PPI}	{EQ, PP, PPI, PO}	{PPI}	{PPI, PO}	{PPI, PO, DR}
PO	{PO}	{PP, PO}	{PPI, PO, DR}	$\mathcal{U}$	{PPI, PO, DR}
DR	{DR}	{PP, PO, DR}	{DR}	{PP, PO, DR}	$\mathcal{U}$

Table 1: RCC5 composition; row R from x to y, column S from y to z. Apart from the EQ row and column, exactly four entries are singletons: $\mathsf{comp}(\mathsf{PP},\mathsf{PP})$, $\mathsf{comp}(\mathsf{PPI},\mathsf{PPI})$, $\mathsf{comp}(\mathsf{PP},\mathsf{DR})$ and $\mathsf{comp}(\mathsf{DR},\mathsf{PPI})$. Every other entry contains PO.

6. A formalization of all of the above in Lean 4, with an executable `Decidable` instance whose only hypothesis is fragment membership (§8).

Relation to the overview, and to an earlier draft. This paper is the focused companion of an overview of the whole $\mathcal{ALCI}_{\mathrm{RCC5}}$ project [15], which records the history, the routes that failed, the open full-logic problem and the methodology. The division into two papers was recommended by a cold review of that overview by GPT-5.6 Sol, which also drafted a focused paper along these lines [20]. We adopted the recommendation. The present text was written afresh from the Lean artifact rather than adapted from the draft; the differences, and the provenance of each part of the result, are recorded in §12.

Scope, and what is not claimed. The result concerns satisfiability of a single concept. It does not cover general TBoxes or ABoxes: internalizing a TBox needs a universal modality, which in this logic is the conjunction of the five universal restrictions, ∀PO included. Because the fragment is not closed under negation, it also does not decide every subsumption: $C \sqsubseteq D$ is decided exactly when $C \sqcap \neg D$ lies in the fragment, that is, when C has no positive ∀PO and no negative ∃PO, and D has no positive ∃PO and no negative ∀PO. No complexity classification is claimed beyond a doubly exponential upper bound (§9), and the literal procedure is far too expensive to run on any but tiny inputs. Decidability of the full logic remains open.

2 Preliminaries

2.1 RCC5 and its composition table

Let $\mathsf{Rel} = \{\mathsf{EQ},\mathsf{PP},\mathsf{PPI},\mathsf{PO},\mathsf{DR}\}$. Converse swaps PP and PPI and fixes the other three. The composition $\mathsf{comp}(R,S)$ lists the relations that can hold from x to z when R holds from x to y and S from y to z; Table 1 gives it, with $\mathcal{U} = \mathsf{Rel}$. The table is the familiar one for nonempty sets under equality, proper inclusion, disjointness and the remaining overlap, and in the formalization it is not an assumption about sets but a definition that is later proved to agree with them (§3.1).

Definition 2.1 (RCC5 frame)**.** An *RCC5 frame* is a nonempty set W with a map $\rho : W \times W \to \mathsf{Rel}$ such that for all $x, y, z \in W$: $\rho(x,x) = \mathsf{EQ}$; $\rho(x,y) = \mathsf{EQ}$ implies $x = y$; $\rho(y,x)$ is the converse of $\rho(x,y)$; and $\rho(x,z) \in \mathsf{comp}(\rho(x,y), \rho(y,z))$.

That ρ is a total function encodes that the five relations are jointly exhaustive and pairwise disjoint; the second condition makes equality *strong*, i.e. identity. These are exactly the general RCC5 structures of Lutz and Wolter [6, §8].

2.2 Concepts and their semantics

Concepts in negation normal form (NNF) are built from concept names A by

$$C, D \ ::= \ \top \mid \bot \mid A \mid \neg A \mid C \sqcap D \mid C \sqcup D \mid \exists R.C \mid \forall R.C, \qquad R \in \mathsf{Rel}.$$

Inverse roles need no separate constructor, since the inverse of an RCC5 relation is again one. An *interpretation* is an RCC5 frame (W, ρ) with an extension $A^{\mathcal{I}} \subseteq W$ for every concept name. Satisfaction $x \models C$ is as usual for the Boolean cases, and

$$x \models \exists R.C \iff \exists y \in W : \ \rho(x, y) = R \text{ and } y \models C,$$
$$x \models \forall R.C \iff \forall y \in W : \ \rho(x, y) = R \Rightarrow y \models C.$$

C is *satisfiable* if $x \models C$ for some element x of some interpretation.

Definition 2.2 (The fragment)**.** C is $\forall\mathsf{PO}$*-free* if no subconcept of C has the form $\forall\mathsf{PO}.D$. For formulas with a general negation constructor, define $\mathrm{Free}^b(F)$ for a polarity $b \in \{+, -\}$ by recursion: negation flips b, the other Boolean connectives keep it, $\forall\mathsf{PO}$ is forbidden under $+$, $\exists\mathsf{PO}$ is forbidden under $-$, and all other modalities are allowed under both. A formula is in the fragment if $\mathrm{Free}^+(F)$ holds.

Example 2.3 (A composition-forced contradiction)**.** $C_{\mathrm{force}} = \exists\mathsf{PP}.(\exists\mathsf{DR}.A) \sqcap \forall\mathsf{DR}.\neg A$ is in the fragment and unsatisfiable: if $x \,\mathsf{PP}\, y$, $y \,\mathsf{DR}\, z$ and $z \models A$, then $\rho(x, z) \in \mathsf{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}$, and $x \models \forall\mathsf{DR}.\neg A$ contradicts $z \models A$. A procedure that checks only the edges it created would accept it.

Example 2.4 (No finite model property)**.** $C_\infty = \exists\mathsf{PP}.\top \sqcap \forall\mathsf{PP}.\exists\mathsf{PP}.\top$ is in the fragment and satisfiable, on the natural numbers with $\rho(m, n) = \mathsf{PP}$ for $m < n$. Every model is infinite: from $x \models C_\infty$ one obtains $x \,\mathsf{PP}\, y_1$, and since $\mathsf{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}$ every later y_{k+1} with $y_k \,\mathsf{PP}\, y_{k+1}$ is again a proper superpart of x, so the universal keeps producing successors, all distinct because PP is irreflexive and transitive. The same happens downwards through disjointness: in $\exists\mathsf{DR}.\forall\mathsf{DR}.\exists\mathsf{PPI}.A$, a disjoint witness forces the root to have a proper part, which is again disjoint from the witness because disjointness is inherited by parts ($\mathsf{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}$), and so on without end.

3 Ordered-disjoint structures

The composition table has a compact reading that the rest of the paper uses throughout: PP is an order, DR is a relation inherited downwards along it, and PO is whatever is left.

Definition 3.1 (Ordered-disjoint structure)**.** An *ordered-disjoint structure* $(N, <, \#)$ consists of a nonempty set N, an irreflexive transitive relation $<$, and a symmetric irreflexive relation $\#$, such that $x < y$ implies $\neg(x \# y)$, and such that, writing $\leq$ for the reflexive closure of $<$,

$$x \# y, \quad x' \leq x, \quad y' \leq y \quad \Longrightarrow \quad x' \# y'.$$

Its *induced labelling* assigns to (x, y) the value EQ if $x = y$, PP if $x < y$, PPI if $y < x$, DR if $x \# y$, and PO otherwise.

The axioms make the five cases mutually exclusive, so the induced labelling is a function.

Theorem 3.2 (Normal form)**.** *(a) The induced labelling of an ordered-disjoint structure is an RCC5 frame.*

(b) Conversely, in every RCC5 frame, $x < y :\iff \rho(x,y) = \mathsf{PP}$ and $x\#y :\iff \rho(x,y) = \mathsf{DR}$ define an ordered-disjoint structure whose induced labelling is ρ.

Proof. (b) Transitivity of $<$ is $\mathsf{comp}(\mathsf{PP},\mathsf{PP}) = \{\mathsf{PP}\}$; irreflexivity of $<$ and $\#$ and their disjointness follow from $\rho(x,x) = \mathsf{EQ}$ and ρ being a function; symmetry of $\#$ is converse-coherence. For inheritance, let $x\#y$ and $x' < x$: then $\rho(x',y) \in \mathsf{comp}(\mathsf{PP},\mathsf{DR}) = \{\mathsf{DR}\}$. If further $y' < y$, then $\rho(x',y') \in \mathsf{comp}(\mathsf{DR},\mathsf{PPI}) = \{\mathsf{DR}\}$. The five values of ρ are read back case by case.

(a) Identity, strong equality and converse-coherence are immediate from the definition. For composition one checks, for each pair (R,S), that every relation outside $\mathsf{comp}(R,S)$ is ruled out by the axioms. The four singleton entries are transitivity and inheritance again. The remaining exclusions are all of one kind; for instance, $\mathsf{comp}(\mathsf{PO},\mathsf{PP}) = \{\mathsf{PP},\mathsf{PO}\}$ excludes DR, because $x\#z$ and $y < z$ would give $x\#y$ by inheritance, contradicting $\rho(x,y) = \mathsf{PO}$, and excludes PPI and EQ, because $z \le x$ and $y < z$ would give $y < x$. The case analysis is routine; it is machine-checked (`odNet_frame`) and has also been tested exhaustively against the table on all structures with at most four elements. □

Remark 3.3 (Two readings of the same shape)**.** The normal form has a precursor in a neighbouring field. A partial order with a symmetric, irreflexive conflict relation inherited along the order is the order skeleton of a prime event structure [7]. With the order reversed—causality running from a region to its parts, since disjointness is inherited downward—DR plays the role of conflict and PO that of concurrency. Region structures need not satisfy the finite-causes axiom of event structures, since an element may have infinitely many superparts. The correspondence was pointed out by GPT-5.6 Sol's review [20]; we do not use it below.

3.1 Representation by sets

For an ordered-disjoint structure $(N,<,\#)$ put

$$\eta(x) \;=\; \{(u,v) \in N \times N \;:\; \neg(u\#v) \text{ and } (u \le x \text{ or } v \le x)\}.$$

The pair (x,x) lies in $\eta(x)$, so every $\eta(x)$ is nonempty.

Lemma 3.4 (Canonical representation)**.** *$\eta(x) \subseteq \eta(y)$ iff $x \le y$; $\eta(x) \cap \eta(y) = \emptyset$ iff $x\#y$; and η is injective.*

Proof. If $x \le y$ then $\eta(x) \subseteq \eta(y)$ by transitivity. Conversely, $(x,x) \in \eta(x) \subseteq \eta(y)$ forces $x \le y$. If $\neg(x\#y)$, the pair (x,y) lies in both sets. If $x\#y$ and (u,v) lay in both, then some coordinate of (u,v) would be $\le x$ and some coordinate $\le y$; by inheritance those coordinates would be in conflict, which is impossible if they coincide (irreflexivity) and contradicts $\neg(u\#v)$ otherwise. Injectivity follows from the first claim and antisymmetry of $\le$. □

The design of η is worth a sentence. The obvious candidate, the down-set of x, loses overlap: two incomparable, non-disjoint elements with nothing below both would get disjoint down-sets. $\eta(x)$ records instead every non-conflicting pair that touches the down-set of x, and the pair (x,y) itself is the common element that keeps overlapping elements overlapping.

For a family of nonempty sets indexed by W, pairwise distinct, let $\rho_{\mathrm{set}}(x,y)$ be EQ, PP, PPI, DR or PO according to whether the two sets are equal, the first is properly included in the second, the converse, they are disjoint, or none of these. Call C *set-satisfiable* if it is satisfied in such an interpretation. This definition mentions no composition table.

Theorem 3.5 (Abstract and set satisfiability coincide)**.** *For every concept C of $\mathcal{ALCI}_{\mathrm{RCC5}}$, with no fragment assumption, C is satisfiable if and only if it is set-satisfiable.*

Proof. A family of sets yields an ordered-disjoint structure (proper inclusion and disjointness), whose induced labelling is ρ_{set}; by Theorem 3.2(a) it is an RCC5 frame. Conversely, read an ordered-disjoint structure off a frame by Theorem 3.2(b) and apply Lemma 3.4; satisfaction transfers because the relation does. □

Remark 3.6 (Prior work)**.** That RCC5 structures can be represented by regions is not new. Lutz and Wolter prove that every general RCC5 structure is isomorphic to a structure of regular closed regions of some topological space, and every countable one to such a structure in $\mathbb{R}^n$, for each $n > 0$ [6, Thm. 23]; passing to interiors gives a representation by sets. Theorem 3.5 is therefore a special case of their result. What is added here is the explicit order-and-disjointness form, the explicit construction η, and proofs that are checked by machine.

4 The cone scheme

Fix an NNF concept C_0. Let $\mathsf{cl}(C_0)$ be the list of its subconcepts, C_0 included, and m its length. The procedure never looks at a model; it works on a finite set of *signatures* determined by C_0 alone.

Types. A *candidate type* is a sublist T of $\mathsf{cl}(C_0)$; there are at most 2^m. A type is a *support type* if $\bot \notin T$, no A and $\neg A$ both occur, $C \sqcap D \in T$ implies $C, D \in T$, $C \sqcup D \in T$ implies $C \in T$ or $D \in T$, and $\forall \mathsf{EQ}.D \in T$ or $\exists \mathsf{EQ}.D \in T$ implies $D \in T$. Types need not be maximal; they record what a point is required to satisfy, not everything it does satisfy. The EQ clauses are local because, with strong equality, the only EQ-neighbour of a point is itself. For a type T and a relation R write

$$\mathsf{B}_R(T) = \{D : \forall R.D \in T\}$$

for the bodies of its universal restrictions along R.

Signatures. A *signature* is a pair $q = (T, S)$ of a candidate type T and a sublist S of candidate types, read as the types that may occur *strictly below* a point of type T in the part-of order. Its *cone* is $\mathsf{cone}(q) = \{T\} \cup S$. A signature is *statically admissible* if T and every member of S are support types and, for every $U \in S$,

$$\mathsf{B}_{\mathsf{PP}}(U) \subseteq T \qquad \text{and} \qquad \mathsf{B}_{\mathsf{PPI}}(T) \subseteq U.$$

The first clause says that a universal over proper superparts, held somewhere below, is honoured here; the second that a universal over proper parts, held here, is honoured below. Let Σ_0 be the set of statically admissible signatures; $|\Sigma_0| \leq 2^m \cdot 2^{2^m}$.

Demands and compatibility. The *demands* of $q = (T, S)$ are the pairs (R, D) with $\exists R.D \in T$ and $R \neq \mathsf{EQ}$. A signature $q' = (T', S')$ is a *compatible target* for the demand (R, D) of q if $D \in T'$ and the condition in Table 2 holds.

The PP row does more than it seems. Together with static admissibility of the target it propagates universals in both directions along the edge: if $\forall \mathsf{PP}.E \in T$ then $T \in \mathsf{cone}(q) \subseteq S'$ forces $E \in T'$, and if $\forall \mathsf{PPI}.E \in T'$ then $E \in T$. The PO row is vacuous on the fragment and matters only for the refutation test of §7.4; an early version of the formalization omitted it, which a cold review caught (§8.4).

Demand	Condition on $q = (T, S)$ and $q' = (T', S')$	Why a model satisfies it
$\exists \mathsf{PP}.D$	$\mathsf{cone}(q) \subseteq S'$	what is at or below x is below y
$\exists \mathsf{PPI}.D$	$\mathsf{cone}(q') \subseteq S$	dually
$\exists \mathsf{DR}.D$	$\mathsf{B}_{\mathsf{DR}}(U) \subseteq V$ and $\mathsf{B}_{\mathsf{DR}}(V) \subseteq U$ for all $U \in \mathsf{cone}(q)$, $V \in \mathsf{cone}(q')$	parts of disjoint regions are disjoint
$\exists \mathsf{PO}.D$	$\mathsf{B}_{\mathsf{PO}}(T) \subseteq T'$ and $\mathsf{B}_{\mathsf{PO}}(T') \subseteq T$	PO is its own converse

Table 2: Transition compatibility. In every row the target must also contain the demanded body D. The third column anticipates the completeness proof, in which x is a point of a model and y its witness.

Elimination. For a set X of signatures let

$$\mathsf{Prune}(X) = \{q \in X : \text{every demand of } q \text{ has a compatible target in } X\}.$$

Prune is reductive ($\mathsf{Prune}(X) \subseteq X$) and monotone ($X \subseteq Y$ implies $\mathsf{Prune}(X) \subseteq \mathsf{Prune}(Y)$), since it only asks that a target *exist*. Put $X_0 = \Sigma_0$ and $X_{n+1} = \mathsf{Prune}(X_n)$, and *accept* C_0 if some $(T, S) \in X_N$ with $N = |\Sigma_0|$ has $C_0 \in T$.

Lemma 4.1 (Fixed point)**.** *X_N is a fixed point of* Prune*. Moreover, every $Z \subseteq \Sigma_0$ with $Z \subseteq \mathsf{Prune}(Z)$ satisfies $Z \subseteq X_n$ for every n.*

Proof. The sequence decreases, and each strict decrease removes a signature, so it is constant from some $n \leq N$ on. For the second claim, $Z \subseteq X_0$, and $Z \subseteq X_n$ gives $Z \subseteq \mathsf{Prune}(Z) \subseteq \mathsf{Prune}(X_n) = X_{n+1}$ by monotonicity. □

Remark 4.2 (Monotonicity is the point)**.** An earlier quasimodel procedure for this logic, since retracted, eliminated types by a rule under which a larger set could be *more* restrictive; its elimination cascaded and rejected satisfiable concepts. Lemma 4.1's second half is exactly what monotonicity buys, and it is the whole of the completeness argument below.

5 Completeness: every model is a post-fixed point

Let $\mathcal{I} = (W, \rho, \cdot^{\mathcal{I}})$ be any interpretation, with no fragment assumption. For $x \in W$ let

$$\mathsf{tp}(x) = \{D \in \mathsf{cl}(C_0) : x \models D\}, \qquad \mathsf{down}(x) = \{\mathsf{tp}(y) : y \in W,\ \rho(y, x) = \mathsf{PP}\},$$

and call $\sigma(x) = (\mathsf{tp}(x), \mathsf{down}(x))$ the *signature of* x. The signatures a model realizes form a finite set, whatever the size of the model, because they are drawn from Σ_0:

Lemma 5.1. *$\sigma(x) \in \Sigma_0$ for every $x \in W$.*

Proof. The support conditions hold because $\mathsf{tp}(x)$ is the set of subconcepts true at x. If $\rho(y, x) = \mathsf{PP}$ and $\forall \mathsf{PP}.D \in \mathsf{tp}(y)$, then $x \models D$; if $\forall \mathsf{PPI}.D \in \mathsf{tp}(x)$, then $y \models D$, since $\rho(x, y) = \mathsf{PPI}$. □

Lemma 5.2 (Witnesses are compatible targets)**.** *If $x \models \exists R.D$ with $R \neq \mathsf{EQ}$, and y is a witness, then $\sigma(y)$ is a compatible target for the demand (R, D) of $\sigma(x)$.*

Proof. $D \in \mathsf{tp}(y)$. For $R = \mathsf{PP}$: x and every proper part of x are proper parts of y, by $\mathsf{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}$, so $\mathsf{cone}(\sigma(x)) \subseteq \mathsf{down}(y)$. The case $R = \mathsf{PPI}$ is dual. For $R = \mathsf{DR}$: if u is x or a proper part of x, and v is y or a proper part of y, then $\rho(u, v) = \mathsf{DR}$ by inheritance (Theorem 3.2(b)), so universals over DR at u hold at v and conversely. For $R = \mathsf{PO}$: $\rho(y, x) = \mathsf{PO}$ as well, so universals over PO cross the edge both ways. □

Theorem 5.3 (Completeness)**.** *If C_0 is satisfiable, then for every n some signature in X_n contains C_0 in its type. In particular C_0 is accepted.*

Proof. Let $Z = \{\sigma(x) : x \in W\}$. By Lemma 5.1, $Z \subseteq \Sigma_0$. By Lemma 5.2, every demand of every $\sigma(x) \in Z$ has a compatible target $\sigma(y) \in Z$, so $Z \subseteq \mathsf{Prune}(Z)$. Lemma 4.1 gives $Z \subseteq X_n$, and $C_0 \in \mathsf{tp}(x)$ for a point x satisfying C_0. □

Nothing here extracts a finite object from the model, bounds a width, or forms a quotient. The finite state space was fixed before the model was mentioned, and the model contributes a single fact: that the signatures it realizes support one another.

6 Soundness: the fresh-occurrence unfolding

Now assume C_0 is $\forall$PO-free, and let $X \subseteq \Sigma_0$ satisfy $X \subseteq \mathsf{Prune}(X)$ and contain $q_0 = (T_0, S_0)$ with $C_0 \in T_0$. We build a model of C_0.

Occurrences. Fix, for each $q \in X$ and each demand of q, the first compatible target in X (one exists because $X \subseteq \mathsf{Prune}(X)$). Number the demands of each signature. An *occurrence* is a finite word over $\mathsf{Rel} \times \mathbb{N}$, the most recent step first. Following a word from q_0, taking the chosen target at each step, assigns it a signature $\mathsf{sig}(w)$; its *label* $\mathsf{lab}(w)$ is the type of $\mathsf{sig}(w)$. The occurrences we keep are the *generated* ones: the empty word, and $(R, i) \cdot w$ whenever w is generated and the i-th demand of $\mathsf{sig}(w)$ is (R, D) for some D. Distinct words are distinct elements, even when they reach the same signature: this is the only sense in which the finite set X describes an infinite model, and no element is ever reused.

The order. The *base steps* are $w \to (\mathsf{PP}, i) \cdot w$ and $(\mathsf{PPI}, i) \cdot w \to w$, between generated occurrences: a PP-child lies above its parent, a PPI-child below it. Let $<$ be the transitive closure of the base steps, and $\leq$ its reflexive closure. Define the *height* $\mathsf{h}(w)$ by adding $+1$ for each PP step in w and -1 for each PPI step. Every base step raises the height by one, so $<$ is irreflexive; it is transitive by definition.

Disjointness. Call $\{w, (\mathsf{DR}, i) \cdot w\}$ a DR *birth*, and set $u \# v$ if $u \leq a$ and $v \leq b$ for some DR birth $\{a, b\}$. This relation is symmetric and inherited downwards by construction. That it is irreflexive and never meets $<$ needs one invariant. The *vertical base* $\mathsf{vb}(w)$ strips the leading PP and PPI steps from w. Base steps preserve it. A DR birth does not: $\mathsf{vb}((\mathsf{DR}, i) \cdot w) = (\mathsf{DR}, i) \cdot w$ is longer than $\mathsf{vb}(w)$, which is a suffix of w. Hence comparable occurrences have equal vertical bases, while $u \# v$ implies $\mathsf{vb}(u) \neq \mathsf{vb}(v)$; so $\#$ is irreflexive and disjoint from $<$.

Lemma 6.1 (The unfolding is a frame)**.** *The generated occurrences with $<$ and $\#$ form an ordered-disjoint structure, so by Theorem 3.2(a) its induced labelling is an RCC5 frame. In it, each birth $(R, i) \cdot w$ stands in relation R to w.*

Proof. The axioms were verified above. PP and PPI births are base steps and DR births are seeds of #. A PO birth changes the vertical base, so its endpoints are incomparable; they are not disjoint either, so the induced labelling gives PO. □

Propagation. Interpret a concept name A as the set of occurrences whose label contains A. Three facts carry the universals.

(P1) *Cones travel up the order.* If $u < v$ then $\mathsf{cone}(\mathsf{sig}(u))$ is contained in the second component of $\mathsf{sig}(v)$. For a base step this is the PP or PPI row of Table 2, and along a chain it composes, since a cone contained in the next spectrum is contained in the next cone. By static admissibility of $\mathsf{sig}(v)$, a $\forall\mathsf{PP}.E$ in $\mathsf{lab}(u)$ puts E into $\mathsf{lab}(v)$, and a $\forall\mathsf{PPI}.E$ in $\mathsf{lab}(v)$ puts E into $\mathsf{lab}(u)$.

(P2) *Disjointness descends from a birth.* If $u\#v$ through the birth $\{a, b\}$, then $\mathsf{lab}(u) \in \mathsf{cone}(\mathsf{sig}(a))$ and $\mathsf{lab}(v) \in \mathsf{cone}(\mathsf{sig}(b))$ by (P1), and the DR row, which relates every pair drawn from the two cones, carries $\forall$DR bodies from $\mathsf{lab}(u)$ to $\mathsf{lab}(v)$ and back.

(P3) *Every demand is served.* If $\exists R.D \in \mathsf{lab}(w)$ with $R \neq \mathsf{EQ}$, the corresponding child of w is generated, stands in relation R to w by Lemma 6.1, and has D in its label.

Lemma 6.2 (Truth lemma)**.** *If C_0 is $\forall$PO-free, then every concept in the label of a generated occurrence is true at that occurrence.*

Proof. By induction on the concept. The Boolean cases, $\bot$, literals and the two EQ modalities follow from the support conditions and strong equality. $\exists R.D$ with $R \neq \mathsf{EQ}$ is (P3). $\forall\mathsf{PP}.E$ and $\forall\mathsf{PPI}.E$ follow from (P1), since every PP- or PPI-neighbour is comparable, and $\forall\mathsf{DR}.E$ from (P2), since every DR-neighbour is related through #. Finally, a label containing $\forall\mathsf{PO}.E$ would put $\forall\mathsf{PO}.E$ into $\mathsf{cl}(C_0)$, which the fragment excludes, so that case does not arise. □

Theorem 6.3 (Soundness)**.** *If C_0 is $\forall$PO-free and some $X \subseteq \Sigma_0$ with $X \subseteq \mathsf{Prune}(X)$ contains a signature whose type contains C_0, then C_0 is satisfiable, in a countable model whose relation is induced by an ordered-disjoint structure.*

Proof. Apply Lemma 6.2 to the empty word; occurrences are finite words over a countable alphabet. □

Remark 6.4 (Where the fragment is used, and what it saves)**.** The hypothesis enters once, in the last case of Lemma 6.2, and the reason is visible in Lemma 6.1. Most PO edges of the unfolding are not births: every pair that is neither comparable nor disjoint is labelled PO, including siblings produced for unrelated demands and occurrences in different branches. The scheme controls the edges it creates, and a $\forall$PO would have to control all the others. Without $\forall$PO these residual edges constrain nothing, so the construction never has to decide jointly how the witnesses of one signature relate to one another. That joint decision is precisely what a reuse-based construction could not make locally (§10).

6.1 The unfolding as a split forest

The shape of this model is not new to the project: it is Wessel's split-forest idea, and both the idea and how much of it the cone scheme keeps deserve stating.

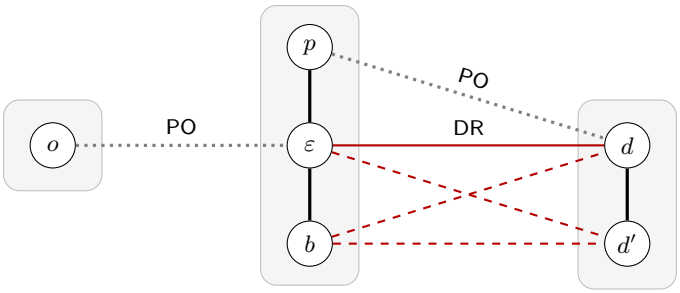


Figure 1: Part of a fresh-occurrence unfolding, drawn as a split forest. The root ε has four demands, served by $p = (\mathsf{PP}, 0)$, $b = (\mathsf{PPI}, 1)$, $d = (\mathsf{DR}, 2)$ and $o = (\mathsf{PO}, 3)$, and d has a proper part $d' = (\mathsf{PPI}, 0) \cdot d$. The shaded boxes are the connected components of the order, each a tree of part-of births (thick): $b < \varepsilon < p$ and $d' < d$. Between components everything is horizontal and read off: DR at the disjointness birth $\{\varepsilon, d\}$ (solid) and, by inheritance, between everything below its two ends (dashed); every other pair, such as (o, ε) and (p, d), is PO (dotted).

The split-forest representation. A model of $\mathcal{ALCI}_{\mathrm{RCC5}}$ is a complete graph, since every two regions stand in some relation. The idea is that it can nevertheless be *presented* as a forest. Take the proper-part order as the skeleton and, wherever a region is demanded as a part by several unrelated superparts, give each demand its own copy instead of letting them share one element; the vertical structure then becomes a forest of trees, one per connected component of the order. Every edge that is not a tree edge relates regions side by side, and is DR or PO, and the idea's second half is that these horizontal edges can be completed afterwards. In the project's earlier routes the completion was justified by the patchwork property of RCC5, the copies were re-identified by a quotient, and the hard part was to bound the interfaces through which horizontal relations cross between branches, so that a finite abstraction of the forest could decide satisfiability [15].

The fresh-occurrence unfolding is a split forest. Occurrences are words, and no two demands ever share a witness: that is the splitting, and it is never undone. The only order edges are part-of births, so each connected component of the order is a subtree of vertical births—its Hasse diagram is exactly that subtree—rooted at the empty word or at a disjointness or overlap birth. The vertical base vb of §6 is the map sending an occurrence to the root of its component. Everything between components is horizontal and is read off rather than chosen: DR for the pairs below the two ends of a disjointness birth, and PO for every other pair (Figure 1). One tree per component, linked horizontally: the split-forest picture, built literally.

What the cone scheme keeps, and what it drops. The two directions of the proof use the idea differently. The soundness direction *builds* a split forest: every model the procedure exhibits has that shape, and the two frame conditions that defeated reuse-based constructions—irreflexivity of the order, and the order never meeting disjointness—hold because it is a forest of vertical trees, in which an order edge cannot leave its component and a disjointness birth opens a new one. The completeness direction does not *assume* a split forest. It takes an arbitrary model, which need not be tree-shaped. What it reads off that model fits the same picture—a cone signature is a point's type together with the types strictly below it, that is, its vertical position, and the transition conditions push cones along PP and PPI and ask nothing structural of PO—but that is a correspondence rather than a documented design step: the cone-scheme plan does not refer to split forests, and a type together with the types below it is a standard invariant for logics over a transitive relation. What

the earlier routes built in order to *obtain* such a forest from a given model is gone. There is no appeal to patchwork, because composition closure follows from the ordered-disjoint axioms (Theorem 3.2); no quotient, because no particular model has to be recovered; and no witness thinning, interface catalogues or width bound, because completeness needs only that a model's signatures support one another (Theorem 5.3).

Why this works on the fragment, and where it stops. The part of the split-forest programme that resisted every earlier attempt was control of the horizontal relations between branches. In the unfolding those relations are never chosen: disjointness is inherited, and everything else defaults to overlap. Without $\forall\mathsf{PO}$ no formula can object to that default, so nothing that crosses between branches needs controlling. With $\forall\mathsf{PO}$ it does—the unsatisfiable concepts C_{bad} and C_{joint} of §7.4 are exactly obligations on such cross-branch pairs—and the interface problem of the original programme returns.

7 Decision theorems

Theorem 7.1 (Decidability, NNF)**.** *For every $\forall\mathsf{PO}$-free NNF concept C_0,*

$$C_0 \text{ is satisfiable} \iff \exists (T,S) \in X_{|\Sigma_0|} :\ C_0 \in T.$$

The right-hand side is a finite computation on lists, so satisfiability of the fragment is decidable.

Proof. Left to right is Theorem 5.3. Right to left: $X_{|\Sigma_0|}$ is a fixed point by Lemma 4.1, so Theorem 6.3 applies. □

7.1 Raw formulas

Normalization is defined by a single recursion carrying a polarity: $\mathrm{nnf}^+(F)$ is intended to be equivalent to F and $\mathrm{nnf}^-(F)$ to $\neg F$, with $\mathrm{nnf}^{\pm}(\neg F) = \mathrm{nnf}^{\mp}(F)$ and nnf^- dualizing each constructor.

Theorem 7.2 (Decidability, raw formulas)**.** *Both polarities of normalization preserve truth pointwise in every interpretation, and $\mathrm{Free}^b(F)$ implies that $\mathrm{nnf}^b(F)$ is $\forall\mathsf{PO}$-free. Hence satisfiability of formulas F with $\mathrm{Free}^+(F)$ is decidable.*

The polarity condition is forced, not chosen: without it, $\neg\exists\mathsf{PO}.A$ would pass a syntactic "no $\forall\mathsf{PO}$" check and normalize to $\forall\mathsf{PO}.\neg A$. The formal decision procedure for raw input is evaluated by the Lean kernel on small inputs, which shows that it computes rather than appealing to excluded middle.

7.2 Sets and regions

Corollary 7.3 (Set semantics)**.** *Set-satisfiability of $\forall\mathsf{PO}$-free concepts is decidable, by the same procedure (Theorems 3.5 and 7.1).*

Corollary 7.4 (Regular closed regions)**.** *For each $n > 0$, satisfiability of $\forall\mathsf{PO}$-free concepts in substructures of the RCC5 structure of regular closed regions of $\mathbb{R}^n$ is decidable, by the same procedure.*

Proof. Lutz and Wolter prove that the logic of general RCC5 structures coincides with that of substructures of $\mathbb{R}^n$ with its regular closed regions [6, Thm. 24]. Directly: the model of Theorem 6.3 is countable, and every countable general RCC5 structure is realized by regular closed regions of $\mathbb{R}^n$ [6, Thm. 23]. □

This does *not* extend to the full structure of *all* regular closed regions of $\mathbb{R}^n$: a universal restriction there ranges over regions that a substructure may omit, and the corresponding logics are different; for RCC5 they are undecidable [6, Cor. 26]. The corollary is also external to the formalization, which proves the set version only.

7.3 An extension: concepts whose ∀PO restrictions are erasable

An external review by Astra 6 (October 2026) [19] proved that the boundary can be pushed past ∀PO-freeness itself, and the proof is now part of the formalization. Let $\mathcal{D}$ be the NNF concepts with no occurrence of ∃PP, ∃PO or ∀DR — but with ∀PO *permitted* — and let $\mathrm{er}(C)$ replace every subformula $\forall\mathsf{PO}.D$ by $\top$.

Theorem 7.5 (PO-erasure [19]; Lean: `erase_equisat`)**.** *For every $C \in \mathcal{D}$, C is satisfiable iff* $\mathrm{er}(C)$ *is. Hence satisfiability on $\mathcal{D}$ is decidable (`decidableSat_dfrag`, `decidableSetSat_dfrag`): erase, then run the procedure of Theorem 7.1 on the* ∀PO*-free result.*

The forward direction is monotonicity. The backward direction builds, from any model of $\mathrm{er}(C)$, a witness forest in which *PO is empty*: only ∃PPI and ∃DR demands generate children, the order is the ancestor relation of the resulting unique-parent forest, and disjointness is *incomparability* — admissible here, and only here, because in such a forest incomparability is downward hereditary (a common descendant of two nodes would place both on one ancestor chain). With no PO pair on the domain, every erased ∀PO is vacuous, so the model satisfies C itself. This is a split forest in the sense of §6.1 with the overlap stripped out entirely.

The extension is a strict syntactic one — $\forall\mathsf{PO}.A \in \mathcal{D}$ is not ∀PO-free (`allpo_dfrag`, `allpo_not_pofree`, `allpo_satisfiable`) — but a calibrated one: inside $\mathcal{D}$ the ∀PO restrictions are provably toothless, which is exactly what the construction shows. Each excluded constructor is necessary. With ∃PO, $\exists\mathsf{PO}.\top \sqcap \forall\mathsf{PO}.\bot$ is unsatisfiable while its erasure is not; the review's witnesses for ∃PP and ∀DR (`wp136`) force two witnesses into a PO relation that a surviving ∀PO then contradicts, so erasure fails there too. The review also proved a normalization theorem behind this: for a *fixed* order, the largest admissible disjointness relates exactly the pairs with no common lower bound, and principal down-sets realize it (Lean: `odMax`, `odMax_largest`, `pdown_subset_iff`, `pdown_disjoint_iff`); its interpretation-level corollary — concepts without ∃PO and ∀DR can be evaluated over principal down-sets — is not formalized and rests on the review's proof and our probe.

7.4 A one-sided test for the full logic

Theorem 5.3 never used the fragment.

Corollary 7.6 (Sound refutation)**.** *For every NNF concept C_0 and every n: if no signature in X_n contains C_0, then C_0 is unsatisfiable.*

The converse fails outside the fragment, and it is instructive to see where. The concept $\forall\mathsf{PO}.A \sqcap \exists\mathsf{PO}.\neg A$ is refuted in the first round: its PO demand needs a target containing both $\neg A$ and, by the PO row, A (the kernel checks the same argument for $\exists\mathsf{PO}.\top \sqcap \forall\mathsf{PO}.\bot$, `cpo_refuted_at_one`). But consider

$$C_{\mathrm{bad}} = \exists\mathsf{PPI}.(\exists\mathsf{PP}.A) \sqcap \forall\mathsf{PP}.\neg A \sqcap \forall\mathsf{PPI}.\neg A \sqcap \neg A \sqcap \forall\mathsf{PO}.\neg A.$$

The root x and the A-point z share a proper part, so $\rho(x,z) \in \mathsf{comp}(\mathsf{PPI},\mathsf{PP}) = \{\mathsf{EQ},\mathsf{PP},\mathsf{PPI},\mathsf{PO}\}$, and the four constraints at the root exclude all four values: C_{bad} is unsatisfiable. Yet a family of three signatures—root, part, and A-point—supports itself under Prune, so the test accepts. The

obligation lives on the pair (x, z), which no demand creates and no row of Table 2 sees. The same happens with an explicit overlap demand in $C_{\text{joint}} = \exists \mathsf{PP}.(\forall \mathsf{PPI}.A \sqcap \forall \mathsf{PO}.A) \sqcap \exists \mathsf{PO}.\neg A$, whose two witnesses stand in a relation from $\mathsf{comp}(\mathsf{PPI}, \mathsf{PO}) = \{\mathsf{PPI}, \mathsf{PO}\}$ and either value forces A onto the $\neg A$-witness. The first of these examples was found by the first cold review of the formalization and the second by GPT-5.6 Sol's review; we verified both acceptances on a transcription of the control layer. And the obvious repair, requiring all witnesses of a signature to be pairwise compatible across PO, is refuted by the satisfiable $C_{\text{sat}} = \exists \mathsf{PP}.(A \sqcap \forall \mathsf{PO}.A) \sqcap \exists \mathsf{PP}.\neg A$, whose two witnesses must be nested, not overlapping. A decision procedure for the full logic has to reason about the relations *among* contemporaneous witnesses, not only along the edges that created them.

8 The formalization

8.1 What is formalized, and how

The development is written in Lean 4 core, without mathlib, and lives in `formal/POFreeLift.lean`, with the forward normal form and the core of η also in `formal/RCC5NormalForm.lean` [21]. The file is cumulative—46,032 lines, most of them earlier routes (§10)—and the cone scheme with its extensions occupies its last few thousand lines. The development log `ASSEMBLY_DESIGN.md`, §§267–297, records it step by step.

The encoding is direct. Relations are an inductive type with composition and converse as functions; concepts are an inductive type in NNF; an interpretation over a carrier type is a domain predicate, a relation function and a valuation; satisfaction is structural recursion; and the frame conditions are a structure over the relation. Satisfiability quantifies over the carrier, so it means satisfiability in an arbitrary frame. Signatures, the elimination and its iteration are lists with Boolean tests, so the decision procedure is a program. Table 3 maps the results of this paper to declarations.

The capstones have these types:

```
decidableSat_cone (C0 :  Concept) (hpo :  POFree C0) :  Decidable (Satisfiable C0)
decidableFSat (F : Formula) (h :  FPOFree true F) : Decidable (FSatisfiable F)
satisfiable_iff_set (C0 :  Concept) :  Satisfiable C0 <-> SetSatisfiable C0
```

8.2 Checking it

The file builds with Lean 4.33.1 and 4.32.0 in under a minute each (`cd formal && elan run leanprover/lean4:v4.33.1 lean POFreeLift.lean`), with no errors, warnings or `sorry`. Newer toolchains (4.34+) deprecate idioms the file uses; build with the pinned versions. The source contains no `axiom` declaration, no `native_decide`, and no `unsafe`, `partial` or `implemented_by`. The capstones depend on the axioms `propext`, `Classical.choice` and `Quot.sound`. Classical reasoning enters where arbitrary interpretations are handled—the type of a point is defined by a classical case distinction on satisfaction—and in the induced labelling of an ordered-disjoint structure, which decides order and disjointness classically; the decision procedure itself computes on lists, and the kernel evaluates it on small inputs. The polarity lemma `pofree_nnfP` and the unsatisfiability of $\exists \mathsf{PO}.\top \sqcap \forall \mathsf{PO}.\bot$ (`cpo_unsat`) use no axioms at all. The repository does not yet pin a toolchain file or provide a minimal build target containing only the dependency closure of the capstones; both are release engineering, not premises of the theorem.

Result	Lean declarations
Normal form, Theorem 3.2	`odNet_frame`, `odOfModel`, `odNet_odOfModel`; `toOrderedDisjoint` (`RCC5NormalForm.lean`)
Representation, Lemma 3.4, Theorem 3.5	`eta_sub_iff`, `eta_disj_iff`, `eta_inj`, `setRel_odNet`, `satisfiable_iff_set`
Scheme and elimination, Lemma 4.1	`sigStatic`, `compatB`, `pruneSig_red`, `pruneSig_mono`, `coneScheme_gfp`, `gfp_greatest`
Completeness, §5	`dkey_compat`, `modelSigs_survives`, `modelSigs_in_gfp`, `coneScheme_complete`
Unfolding, Lemma 6.1	`Gen`, `gOD`, `unfInterp_rcc5`, `unf_edge_po`
Propagation and truth, Lemma 6.2	`allPP_gLt`, `allPPI_gLt`, `allDR_gDisj`, `gchild_serves`, `unf_truth`
Soundness and decision, Theorems 6.3, 7.1	`coneScheme_sound`, `coneScheme_correct_at`, `decidableSat_cone`
Raw formulas, Theorem 7.2	`nnfP_correct`, `pofree_nnfP`, `decidableFSat`
Set semantics	`decidableSetSat`
Refutation, Corollary 7.6	`coneScheme_unsat_full`, `cpo_refuted_at_one`
PO-erasure, Theorem 7.5	`DFrag`, `eraseAllPo`, `forest_truth`, `erase_equisat`, `decidableSat_dfrag`, `decidableSetSat_dfrag`
Maximal disjointness (order level), §7.3	`odMax`, `odMax_largest`, `pdown_subset_iff`, `pdown_disjoint_iff`, `pdown_inj`

Table 3: Results and their Lean declarations, all in `formal/POFreeLift.lean` unless marked.

8.3 What machine checking does and does not settle

The kernel guarantees that the theorems follow from the definitions. It cannot guarantee that the definitions mean what this paper says they mean, and three things in the paper are not formal theorems: the absence of finite models in Example 2.4 (only satisfiability of C_∞ is formalized), the complexity bound of §9, and the transfer to regular closed regions in Corollary 7.4, which cites Lutz and Wolter. Definitional adequacy is what the reviews below were aimed at, and the equivalence with set semantics was proved partly so that it need not rest on anyone's reading of the frame conditions.

8.4 Reviews

The formalization was reviewed by three fresh model sessions with no access to its development history, each targeting a different part, and later audited at source level by a fourth [20].

- The first built the file under two Lean versions, re-derived the composition table from set semantics, compared a transcription of the control layer with exhaustive search over small models on 4,000 random fragment concepts, and found two comments that misstated where the fragment hypothesis is used and an enumeration of types that was larger than necessary by an exponential.
- The second attacked completeness, ran the elimination on 6,268 concepts against exhaustive model search without rejecting a satisfiable one, and found that the PO row of Table 2 had been omitted. Restoring it left the fragment result untouched—the row is vacuous there—and made the refutation test of §7.4 strictly stronger.

- The third attacked soundness, running the unfolding on labels taken from actual greatest fixed points: 2,157 accepted concepts, 6,464 unfoldings and 10,066 occurrences, with no violation of the frame conditions or of the truth lemma. Its findings concerned the project's own test scripts and prose.
- The fourth, by GPT-5.6 Sol, located every capstone of the decision chain in the source and found no hidden hypothesis; its findings concerned how the companion overview described the result.
- A fifth, by Astra 6 (October 2026) [19], audited the argument of this paper and its inspected Lean dependencies and found *no gap* — the first review with nothing to correct — and contributed the extension of §7.3, which we verified (`wp136`) and then formalized.

No review found a counterexample to the theorem.

9 Complexity and practicality

With $m = |\mathsf{cl}(C_0)|$ there are at most 2^m candidate types and $|\Sigma_0| \leq 2^m \cdot 2^{2^m}$ signatures. The procedure performs at most $|\Sigma_0|$ rounds, and a round compares every signature and demand with every candidate target, each comparison being polynomial in m and 2^m. The running time is therefore bounded by $2^{2^{O(m)}}$, doubly exponential in the size of C_0. This bound is not formalized, and no lower bound beyond what the fragment inherits is established here.

The bound describes the literal procedure, and the literal procedure is impractical: it materializes the power set of the type space, and in the project's measurements it can be run only on concepts with a handful of subconcepts. A symbolic implementation—antichains of spectra, decision diagrams, or a SAT encoding of the elimination—is the obvious next step, but it would need its own correctness proof. Merging spectra that agree on the currently relevant bodies is exactly the kind of identification that this construction avoids, and it could reintroduce the problems it removed.

10 Why reuse failed, and what ∀PO breaks

The cone scheme was not the first attempt to certify this fragment. The fragment was isolated in April 2026 by a two-tier quotient construction [16] and re-derived twice by other arguments [15], but formalizing it on those lines required a finite set of nodes, reused by blocking, with witnesses borrowed from one node for another. Four successive disciplines for that reuse were refuted by exact finite countermodels: a per-edge safety test (two individually safe fallback edges formed a four-cycle), a class representative chosen by type (it dragged one member's disjointness witness under another member), a key computed from an occurrence's downward cone (the obstruction was horizontal, so the key could not separate the occurrences), and repeat-free selection.

The failures had one cause. Reusing a node imposes a global condition on the object being built—the order induced by the reused edges must be acyclic, and the relations those edges force must be jointly realizable—and each discipline was a local test, on an edge, a node or a type. Blocking in the usual sense is the same local test: two occurrences with equal types, so let one stand for the other. Equal types cannot show whether the substitution closes a cycle in the order or places a disjointness witness below a node it must be disjoint from. The cone scheme does not test more carefully; it removes the obligation. Nothing is reused, so the order is irreflexive because heights increase, and disjointness never meets the order because a DR birth changes the vertical

base. GPT-5.6 Sol proposed this change of architecture when asked for an alternative to repairing the reuse route [18].

What $\forall\mathsf{PO}$ breaks is then easy to name. The unfolding labels every residual pair PO, and without $\forall\mathsf{PO}$ no residual pair is constrained. With $\forall\mathsf{PO}$, the relations among contemporaneous witnesses matter, as C_{bad} and C_{joint} show, and a construction for the full logic has to choose them jointly. Three directions look promising: control states that carry a complete RCC5 matrix among representatives of simultaneously live demands, whose completeness is where a bound on joint width would have to be proved; regular or automatic presentations of models; and a recasting of ordered-disjoint structures as a single strict order on two copies of the domain, which brings the problem close to two-variable logic with one transitive relation—whose general satisfiability problem is itself open [11, 13]—and adds an involution that keeps the copies synchronized. The last two were suggested by GPT-5.6 Sol's review [20]. Any candidate must reject C_{bad} and accept C_{sat}.

11 Related work

Spatial description logics and modal logics of regions. Cohn proposed modal and non-modal qualitative spatial logics [2]. Wessel introduced the $\mathcal{ALCI}_{\mathrm{RCC}}$ family, settled its coarser members, and left the RCC5 and RCC8 cases open [14]; the same report canvassed restricted fragments that regain the finite model property, which the present fragment does not. Lutz and Wolter [6] introduced modal logics of topological relations. For RCC5 they proved the representation and logic-equivalence results used above (Theorems 23 and 24), recursive enumerability of the logics of general structures (Theorem 28), and undecidability for classes of structures closed under certain suprema (Theorem 25); they left the general case open. They also remarked that dropping PO has been argued to leave a useful formalism for geographic applications, pointing to work of Shapirovsky and Shehtman on a single inclusion modality [9]; that remark concerns usefulness and carries no decidability argument. Our fragment keeps PO and $\exists\mathsf{PO}$ and removes only the universal. So far as we can determine, no decidability result for a PO-restricted RCC5 language has been published.

Constraint reasoning. Consistency of finite networks of RCC relations is the subject of qualitative constraint reasoning; Renz and Nebel analysed its complexity for RCC8 [8]. For atomic RCC5 networks with strong equality, path consistency implies consistency, which is Theorem 3.2(b) together with Lemma 3.4 read for finite structures. Concept satisfiability adds unary predicates, arbitrary nesting and infinite models, so constraint consistency is an ingredient here, not a decision procedure.

Concrete domains. Description logics with concrete domains [5, 1] attach spatial values to abstract objects through features. The closest of them, $\mathcal{ALCRP}(D)$, forms roles from concrete predicates and so quantifies over spatial relations; it is undecidable in general and decidable under syntactic restrictions [4]. It stays two-sorted: equal extensions need not be equal objects, and objects without the feature stand in no spatial relation.

Event structures and two-variable logic. The ordered-disjoint normal form matches the order and conflict skeleton of prime event structures [7]. For two-variable first-order logic with one transitive relation, decidability was claimed [10] and then withdrawn [11]: it holds for transitively guarded witnesses and for finite satisfiability [12], and is open in general, also for a single partial order [13]. The unary case of that open problem is a syntactic fragment of $\mathcal{ALCI}_{\mathrm{RCC5}}$ lying outside

the one decided here (`formal/PosetBridge.lean`, `poset_bridge`), so the synchronized two-copy encoding of §10 would at best relate two open problems.

Formalization. The proof is checked in Lean 4 [3]. Its cumulative development, including the refuted routes, is public.

12 Provenance and contributions

The result was produced by a long human–model collaboration, and the record of who did what is public [22]. We state it here for this result specifically.

- **Michael Wessel** posed the problem in 2002/2003, supplied the split-forest representation—vertical nesting plus horizontal completion—whose shape the unfolding builds (§6.1), and the kernel idea from which the fragment was first isolated. He directed the work throughout, stopped the repair of the reuse architecture when it began to cycle, asked a fresh model for an alternative, and decided on the split into two papers.
- **Claude** (Anthropic) built the two-tier quotient construction that first isolated the fragment (April 2026). Claude Fable 5 formalized the normal form and the core of η (July 2026); Claude Opus 4.8 formalized the logic layer on which the truth lemma rests (July 2026); Claude Opus 5 formalized the cone scheme, the normalization of raw formulas and the equivalence with set semantics (August 2026), and wrote this paper.
- **GPT-5.4 Pro** (OpenAI) reviewed the two-tier construction and exposed the overlap gap that located the fragment's boundary. **GPT-5.5** re-derived the fragment independently in a cold attack on the open problem, and its regular-cover analysis produced the ordered-disjoint normal form. **GPT-5.6 Pro** gave the canonical set representation η [17].
- **GPT-5.6 Sol** (OpenAI) designed the cone scheme: retire node reuse, keep the finite scheme only as a control graph, and unfold every non-`EQ` demand to a fresh occurrence [18]. Later, in a cold review of the overview, it recommended separating this result into a focused paper and wrote a draft of one [20].
- **Astra 6** audited the argument (October 2026, finding no gap) and proved the PO-erasure extension and the maximal-disjointness normalization in prose [19]; this project verified both (`wp136`) and carried the erasure theorem and the order-level normalization lemmas into the Lean artifact (§7.3).
- **Three fresh model sessions** reviewed the formalization cold (§8.4); their findings are incorporated.

How this text relates to the draft. GPT-5.6 Sol's draft is archived unmodified with its review [20]. The recommendation to write a focused paper, and its outline—normal form, scheme, completeness, soundness, decision theorems, formalization, open problem—are that review's. The text is not: it was written anew from the Lean sources, and it departs from the draft where our own checks led elsewhere. It credits Lutz and Wolter's Theorem 23 for the set representation, which the draft listed among its contributions; it marks which statements are cited or argued in prose rather than checked by the kernel; it adds the boundary examples of §7.4, with their status verified against the control layer; and it records provenance in full.

Artifacts. The repository is https://github.com/lambdamikel/alcircc5. The formal files are `formal/POFreeLift.lean` and `formal/RCC5NormalForm.lean`. This paper, the overview, and the Lean artifact are frozen at git tag `release-2026-09-14` (moved 2026-10-08 to include the erasure extension); relative to `certified-fragment-2026-08-29` the Lean sources add §§299–300 and differ otherwise only in the spelling of comments. Cite the tag or its commit hash rather than the `master` branch. The review and draft are in `papers/gpt-5.6-latest-overview-paper-review-and-recommendation/`, and the overview is `papers/overview_arxiv.tex`.

AI disclosure. The mathematics, the formalization and this text were produced by the AI systems named above under the direction of the human author, who is responsible for the paper. The results are machine-checked where stated and have not yet been reviewed by human experts in description logic or qualitative spatial reasoning; we would welcome that review.

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